Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[18AU52]**

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

HEAT AND MASS TRANSFER

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.
- **iii.** Use of Standard HMT data book and Steam tables are permitted

1. a. Present your understanding on thermal conductivity and what is the role of thermal conductivity in heat transfer **04**

b. A cold storage room made of 220mm brick on the outside, 90mm plastic wool and finally 16mm wood on the inside. The outside and inside temperatures are 25°C and -3°C respectively. The inside and outside heat transfer coefficients are 30 and 11W/m^{2}C respectively. The thermal conductivities of the brick foam and wood are 0.99, 0.022 and 0.17W/mK respectively. Determine the 1) rate of heat removal from the refrigerator if area is 85m^2 2) temperature inside surface of the brick. **10**

2. a. A hot plate 1.2m wide and 0.35m height at 115°C is exposed to the still air at 25°C. Calculate the following: **14**

- **i.** Maximum velocity at 180mm from the leading edge of the plate.
- **ii.** The boundary layer thickness at 180mm from the leading edge.
- **iii.** Local heat transfer coefficient at 180mm from the leading edge.
- **iv.** Average heat transfer coefficient over the surface of the plate
- **v.** Total mass flow through the boundary.
- **vi.** Heat loss from the plate.

3. a. Discuss about the Specular and diffuse reflections **04**

b. A black body of 1200cm^2 emits radiation at 1000K , Calculate the i) Total rate of energy emission ii) intensity of normal radiation iii) wavelength of maximum monochromatic emissive power iv) intensity of radiation at an angle of 60° to the normal plane. **10**

OR

4. a. Define the following: **06**

- **i.** Reciprocity theorem
- **ii.** Stefan-Boltzmann Law
- **iii.** Wein's Displacement law

	b.	Calculate the heat exchange by radiation between two long concentric cylinders having 60mm inner radius and 120mm outside radius. The inner cylinder is maintained at 130°C with the emissivity of 0.6 and the outer cylinder is maintained at 30° with the emissivity of 0.5	08
5.	a.	Give the purpose and functionality of Heat exchanger? Classify the types of heat exchangers	06
	b.	Derive Logarithmic Mean Temperature Difference for the Counter flow heat exchanger	08
		OR	
6.	a.	In a counter flow double pipe heat exchanger water is heated from 25°C to 65°C by an oil with a specific heat of 1.45kJ/kgK and a mass flow rate of 0.9kg/s. The oil is cooled from 230°C to 160° C with an overall heat transfer coefficient of 420W/m^{2}C. Calculate: 1) Heat transferred 2) mass flow rate of water 3) Surface area for the heat exchanger.	07
	b.	Water is boiled at the rate of 24kg/s in a copper pan, 300mm diameter, at atmospheric pressure. Assuming nucleate boiling conditions, calculate the bottom surface temperature of the pan.	07
7.	a.	Define the Fick's law of diffusion. Classify the types of diffusion mass transfer	04
	b.	The molecular weight of the two components A and B of a gas mixture are 24 and 28 respectively. The molecular weight of the gas mixture is found to be 30. If the mass concentration of the mixture is 1.2kg/m^3, determine the i) molar and mass concentrations of the components ii) Molar fractions iii) Mass fractions iv) total pressure if the mixture temp is 290K.	10
		OR	
8.	a.	Estimate the diffusion rate of water from the bottom of a test tube of 25mm diameter and 35mm long in to dry air at 25°c. Take mass rate of evaporation water to air 0.28x10^{-4}m^2/s.	06
	b.	Discuss about the analogy between the heat and mass transfer	08